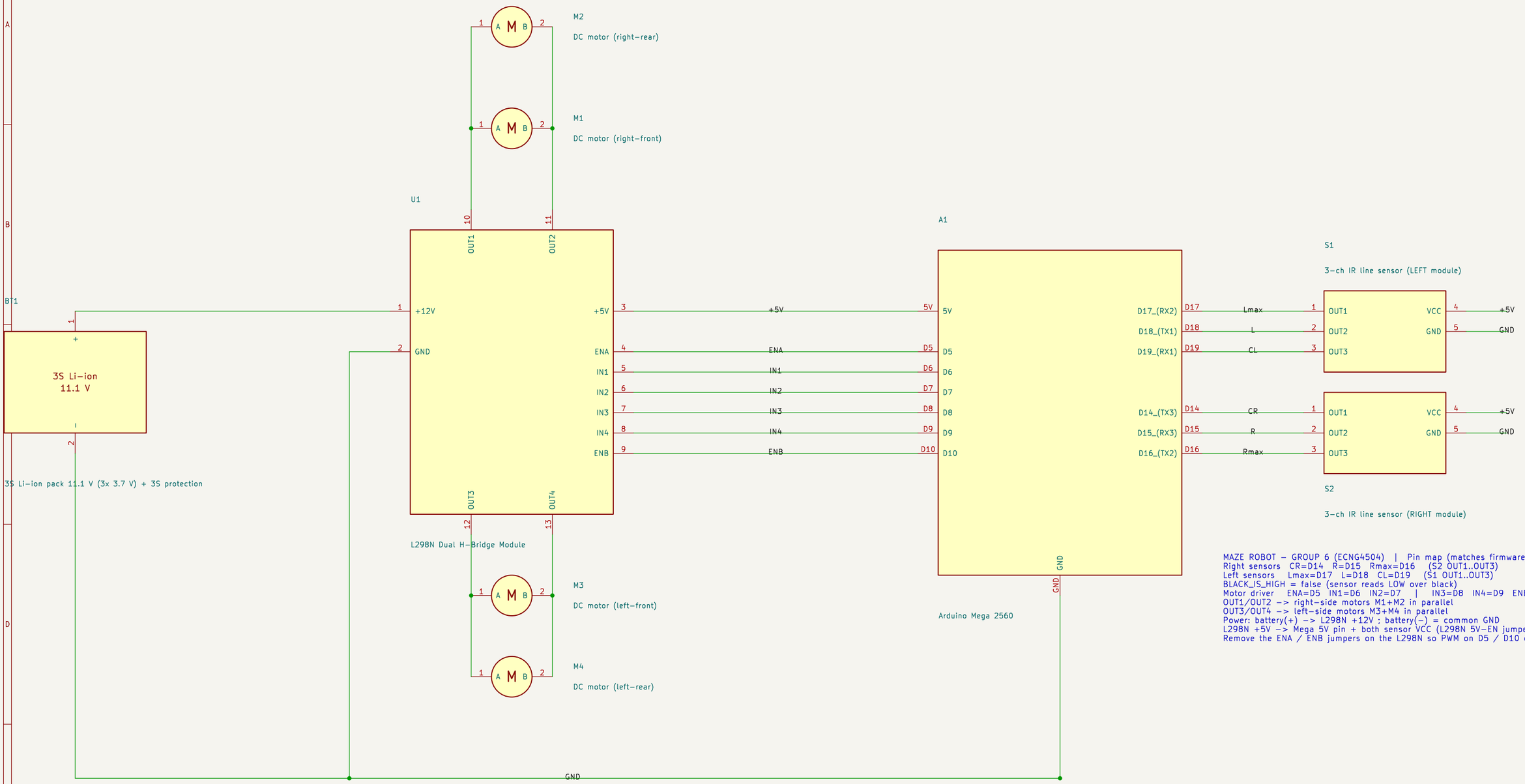


MAZE ROBOT – GROUP 6 (ECNG4504) | Pin map (matches firmware)
Right sensors CR=D14 R=D15 Rmax=D16 (S2 OUT1..OUT3)
Left sensors Lmax=D17 L=D18 CL=D19 (S1 OUT1..OUT3)
BLACK_IS_HIGH = false (sensor reads LOW over black)
Motor driver ENA=D5 IN1=D6 IN2=D7 | IN3=D8 IN4=D9 ENB=D10
OUT1/OUT2 -> right-side motors M1+M2 in parallel
OUT3/OUT4 -> left-side motors M3+M4 in parallel
Power: battery(+) -> L298N +12V ; battery(−) = common GND
L298N +5V -> Mega 5V pin + both sensor VCC (L298N 5V-EN jumper ON)
Remove the ENA / ENB jumpers on the L298N so PWM on D5 / D10 controls speed

The American University in Cairo – ECNG4504 – Group 6

Sheet: /
File: maze_robot.kicad_sch

Title: Maze Robot – Power, Motor Driver and Sensor Schematic

Size: A3	Date: 2026-10-18	**Rev: A**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 1/1